Syrian Private University

Faculty of Artificial Intelligence Engineering

Department of Artificial Intelligence Engineering and Data Science

Speech-to-Sign Language System

(Graduation Project 2) prepared to obtain the degree of B.Sc.
in the Department of Artificial Intelligence Engineering and Data Science

Authors:

Dana Mahmoud Alhaffar

Supervised by:

Dr.Assef Jafar

Eng.Zienab Albaghdadi

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SUPERVISOR CERTIFICATION

I certify that the preparation of this project, entitled "Speech-to-Sign Language Translation System," prepared by "Students Names," was carried out under my supervision at the Faculty of Artificial Intelligence Engineering in partial fulfillment of the requirements for the degree of B.Sc. in the Department of Artificial Intelligence Engineering and Data Science.

Name: _____

Date: _____

Signature: _____

Abstract

The development of accessible communication technologies is an important research area in artificial intelligence, particularly in applications that aim to reduce communication barriers between hearing individuals and Deaf or hard-of-hearing individuals. This project presents a prototype system that converts spoken English into a visual representation of American Sign Language (ASL). The proposed system combines automatic speech recognition, natural language processing, and video processing techniques to generate a sequence of sign-language video clips corresponding to a spoken sentence.

The system first processes an audio input and converts it into text using the Whisper-based speech recognition model. The recognized text is then cleaned and analyzed using natural language processing techniques. A rule-based text-to-gloss component transforms the sentence into a sequence of sign-related keywords, which are subsequently matched against a video bank constructed from the ASL Citizen dataset. The selected video clips are processed and combined into a final output video. When a suitable sign video is unavailable, the system can use a fallback representation to indicate the missing sign.

The main objective is to investigate a practical, modular approach to speech-to-sign video generation using existing speech recognition models, linguistic processing, and sign-language video resources. The prototype also provides a foundation for future improvements, including more accurate gloss generation, improved video alignment, and more natural sign-language production.

Keywords: Sign Language, American Sign Language, Speech Recognition, Whisper, Natural Language Processing, Gloss Generation, ASL Citizen, Video Processing, Accessibility.

Chapter 1: Introduction

1.1 Introduction

Communication is essential for social interaction, education, employment, and access to information. However, communication barriers may arise when hearing individuals and Deaf or hard-of-hearing individuals do not share a common language. Sign languages provide a visual means of communication through hand movements, facial expressions, body posture, and spatial information. Despite their importance, communication between sign-language users and people unfamiliar with sign language can remain challenging.

Artificial intelligence offers several approaches to improving accessibility, including automatic speech recognition, natural language processing, machine translation, and computer vision. These technologies can be combined to develop systems that convert information from one modality into another. In particular, a speech-to-sign system aims to transform spoken language into a visual sequence that represents the meaning of the original message using sign language.

This project focuses on developing a prototype that converts spoken English into ASL video output. Rather than training a complete sign-language generation model from scratch, the proposed approach uses existing speech recognition technology and a collection of sign-language videos. The spoken input is first transcribed into text. The resulting sentence is then processed to extract relevant keywords or glosses, which are matched with available sign videos. Finally, the selected clips are processed and concatenated to produce a single output video.

The project is organized as a modular pipeline, allowing each stage to be developed, tested, and improved independently. This design makes it possible to identify errors in speech recognition, linguistic processing, sign matching, and video generation separately. Nevertheless, because the prototype relies on individual video clips, the resulting sequence may not fully reproduce the grammar, transitions, facial expressions, and natural flow of continuous ASL. These limitations are considered part of the research problem and provide directions for future development.

The project therefore explores the feasibility of combining existing artificial intelligence tools and sign-language resources to create an accessible prototype for converting spoken English into a visual sequence of ASL signs.

1.2 Scientific Problem

The scientific problem addressed by this project is the difficulty of automatically converting spoken English into a meaningful and visually understandable sequence of American Sign Language signs.

Although automatic speech recognition systems can convert speech into text with considerable effectiveness, the recognized text cannot simply be translated into sign language by replacing every English word with a corresponding sign. Sign languages have their own linguistic structures and grammatical rules, which may differ significantly from those of spoken languages. Furthermore, a suitable video may not exist for every word in a sentence, and combining isolated sign clips can produce discontinuities in movement and meaning.

The problem can be divided into the following subproblems:

1. Speech recognition: Converting an audio signal into an accurate English transcription despite variations in pronunciation, speaking speed, and background noise.

2. Text processing and gloss generation: Identifying the important linguistic elements of a sentence and transforming them into a sequence suitable for sign retrieval.

3. Sign-video matching: Matching the generated glosses with the correct sign labels and corresponding video clips in the available dataset.

4. Video composition: Combining the retrieved clips into a single video while maintaining appropriate ordering, timing, and visual consistency.

5. Handling missing signs: Managing words for which no suitable sign video is available in the current video bank.

The central research question is:

How can automatic speech recognition, rule-based natural language processing, and a sign-language video dataset be combined to generate a coherent visual sequence of ASL signs from spoken English?

The project investigates this question through the implementation and evaluation of a modular prototype. Its success depends not only on the accuracy of speech recognition but also on the quality of the gloss sequence, the coverage of the sign-video bank, and the coherence of the final video.

1.3 Research Objectives:

The main objective of this project is to design and implement a prototype that converts spoken English into a sequence of ASL video clips using artificial intelligence and video-processing techniques.

The specific objectives are:

- To develop an audio-processing pipeline that accepts spoken English as input.
- To use a pretrained Whisper-based model for automatic speech recognition.
- To clean and segment the recognized text using natural language processing techniques.
- To implement a text-to-gloss component that converts English sentences into sign-related keywords.
- To construct or use a searchable video bank based on available ASL sign-language data.
- To match generated glosses with their corresponding sign videos.
- To process and concatenate the selected clips into a single output video.
- To handle missing signs through a fallback mechanism.
- To evaluate the prototype in terms of transcription quality, gloss generation, sign coverage, processing time, and output-video quality.
- To identify the limitations of the current approach and establish directions for future improvement.

The project is intended as a prototype and a basis for further research, rather than a complete replacement for a qualified sign-language interpreter.

.1.4 state of the art :

Sign-language technology has developed through several research directions, including isolated sign recognition, continuous sign-language recognition, sign-language translation, and sign-language production. These areas are related, but they address different problems.

5.1. Sign-Language Recognition and Translation

Early sign-language recognition systems focused on identifying individual signs from images or video sequences. More advanced systems attempt to recognize continuous signing or translate sign-language video into spoken-language text. Recent approaches use deep learning, recurrent neural networks, and Transformer-based architectures to model the temporal and linguistic relationships between signs. However, sign-language technology remains challenging because of differences in grammar, signing styles, available datasets, and the visual information conveyed by facial expressions and body movements.

It is important to distinguish sign-language recognition from sign-language production. Recognition interprets signs performed by a person, whereas production generates signs from text or another input modality. The present project addresses the production direction: it starts with spoken English and generates visual sign-language output.

5.2. Gloss-Based Translation

A gloss is a written label used to represent a sign or a meaningful unit in sign language. Gloss-based approaches use an intermediate representation between a source sentence and the target sign sequence. This representation can simplify the process of matching linguistic units to available signs.

5.3. ASL Citizen Dataset

The ASL Citizen dataset was introduced by Desai et al. in 2023 to support research on isolated American Sign Language recognition and sign-dictionary retrieval. The published dataset contains 83,399 videos representing 2,731 distinct signs, recorded by 52 signers in varied environments.

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In this project, ASL Citizen is used as a source of sign videos from which a searchable video bank can be constructed. Each available clip is associated with a sign label or gloss, allowing the system to retrieve candidate clips for words or glosses generated from the input sentence.

However, ASL Citizen was designed primarily for isolated sign recognition and dictionary retrieval, not specifically for generating fluent, continuous sentences from speech. Consequently, using its videos to construct a sentence requires additional processing and does not automatically guarantee grammatical ASL output.

5.4. Automatic Speech Recognition

Automatic speech recognition (ASR) converts spoken language into text. The Whisper family of models provides pretrained speech recognition capabilities that can be integrated into applications without training an ASR model from scratch.

The current implementation uses faster-whisper with the small.en model to transcribe English audio. The audio is converted to a suitable waveform format, and the model produces text segments that are combined into a transcription. This transcription is then passed to the text-processing stage.

Using a pretrained model reduces the need for a dedicated speech dataset and extensive model training. Nevertheless, recognition errors can propagate through the subsequent

stages: an incorrectly recognized word may result in an incorrect gloss or an unavailable sign clip.

5.5. Position of the Proposed System

The proposed system combines existing techniques into a practical prototype rather than introducing a new end-to-end neural network.

6.1. Proposed Solution

The proposed solution is a modular speech-to-sign video generation pipeline. It uses a pretrained speech recognition model, a text-processing module, a gloss-generation component, a sign-video lookup mechanism, and a video-processing module.

The system operates as follows:

1. Audio input: The user provides an English audio recording.
2. Audio preprocessing: The audio is loaded and converted into a suitable format, including mono-channel audio at a sampling rate of 16 kHz.
3. Speech recognition: The Whisper-based model transcribes the audio into English text.
4. Text cleaning and segmentation: The transcription is cleaned, unnecessary patterns are removed, and sentences are identified using NLP tools.
5. Gloss generation: The text-to-gloss component transforms the sentence into a sequence of sign-related labels using linguistic rules and text-processing functions.
6. Sign retrieval: Each generated gloss is searched for in the video bank and its associated lookup information.
7. Missing-sign handling: If no suitable video is available, the system uses the implemented fallback representation.
8. Video processing: The selected clips are processed as necessary, ordered, and concatenated.

9. Output generation: The system produces a final video representing the input sentence as a sequence of sign clips.